

WR-VI

The Aquarium Bounds: Physical Constraints as Boundaries of World Space

Constraint-conditioned transition completion
Mathematical draft v0.2

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Abstract

Responses identify a transition law only relative to a declared admissible class. WR-VI adds an explicit constraint layer to the experimental completion interface of WR-V and asks what a boundary of that class actually certifies. We separate observed responses, structural restrictions and externally justified hard bounds. A finite classical construction couples transition rows through conservation support and detailed balance. A three-state reversible benchmark has identical calibrated mean responses for an entire interval of transition matrices. A bound on stationary transition activity then produces an empty class, a singleton or a nontrivial interval, with an exact threshold and an elementary infeasibility certificate. The selected matrix at the threshold is reducible: an additional irreducibility requirement removes it and changes the threshold's attainability. Positive response tolerance destroys this exact singleton. When the activity cap is uncertain, universal admissibility and admissibility for at least one cap have different quantifiers; a unique universally admissible candidate does not establish unique inference about the actual unknown cap. These are self-contained finite-model calculations using established Markov-chain and convex-feasibility tools. No physical constant, cosmological horizon or boundary of reality is derived.

Keywords: physical admissibility; reversible transitions; activity bound; constrained completion; uncertain constraints; model boundary.

Parallel EN/RU texts share all result and equation numbers. REVIEWABLE_DRAFT; formal external peer-review status unverified; no WR-VI DOI. Content CC BY 4.0; code MIT.

Contents

1	The Aquarium question and the new obligation	3
2	Separate data, restrictions and their justification	3
3	A finite conservation and reversibility interface	3
4	Auditable linear feasibility certificates	4
5	Three states: a complete activity-threshold calculation	5
6	Response tolerance and uncertain caps	6
7	From a transition boundary to a physical boundary	7
8	Provenance, failure criteria and programme handoff	7

1 The Aquarium question and the new obligation

The founding WREH programme uses “Aquarium Bounds” for candidate restrictions on the space of admissible worlds, without positing a surrounding space or an observer outside the Universe. A mathematical boundary can arise because an adopted carrier, constitutive assumption or response fit ceases to be admissible. Establishing a boundary of physical realizability is a stronger obligation. The present draft addresses the former precisely and records what would be needed for the latter.

WR-I supplies response equivalence and identifiable quantities [1]; WR-II distinguishes finite obstruction from failure of full global completion [2]; WR-III supplies geometry of nonempty response fibres [3]; WR-IV distinguishes full realizations from history projections [4]. WR-V declares finite outcome-plus-successor instruments and calibrated transition probes [5]. Its Section 9 hands off the question of emptiness, uniqueness or multiplicity after independently justified admissibility restrictions are imposed; equation (1) below makes that finite constraint interface explicit. Here the new obligation is to construct restrictions that couple instrument rows and to distinguish selection by measured responses from selection that also uses those restrictions. Generic quotient, rank, gluing and smoothing theorems remain upstream results.

Our principal carrier is a finite classical transition matrix. It is a projected description, not a complete world. One experiment step is specified without assigning it a physical duration. The benchmark’s activity is a dimensionless probability of changing a label under a fixed distribution, not energy, entropy production, dissipation or a rate per second. Conservation and equilibrium assumptions require separate justification in any application. No causal speed limit, quantum restriction or gravitational boundary is proved.

2 Separate data, restrictions and their justification

For a declared finite instrument carrier $\mathcal{M}$, let $R : \mathcal{M} \rightarrow \mathcal{Y}$ be a calibrated response map. Data y and a deterministic tolerance $\varepsilon \geq 0$ define $\mathcal{F}_\varepsilon(y) = \{M \in \mathcal{M} : \|R(M) - y\|_\infty \leq \varepsilon\}$. Declare a structural subset $\mathcal{S} \subseteq \mathcal{M}$ and a family of hard-bound subsets $\mathcal{A}_\theta$. The constraint-conditioned completion class is

$$\mathcal{C}_{\theta,\varepsilon}(y) = \mathcal{F}_\varepsilon(y) \cap \mathcal{S} \cap \mathcal{A}_\theta. \quad (1)$$

The subscript θ names supplied model information. It is neither inferred by writing (1) nor a physical time coordinate. A constraint may itself be estimated experimentally, but that separate data source and its uncertainty must be stated. “Physical admissibility” names an application obligation; the set construction alone cannot discharge it.

Definition 2.1 (Constraint record). *A restriction is accompanied by its carrier and units, algebraic condition, physical/model status, source of justification, uncertainty set and allowed relaxation. Absence of any required part is an open obligation, not permission to infer a universal physical law.*

Three diagnostics must be kept apart. If $\mathcal{F}_\varepsilon(y)$ is empty, the data already fail on the declared carrier. If it is nonempty but (1) is empty, at least one added restriction conflicts with the compatible class. If the intersection is a singleton, uniqueness holds only conditional on every used restriction. Dropping a bound while keeping the carrier fixed tests its contribution; changing the carrier is a different model comparison. Such finite conflicts are finite obstructions, not the full-finite defect of WR-II.

A boundary must name its ambient space and topology. Below, “threshold” means a change of feasibility as a scalar cap varies with exact data fixed. We do not promote that threshold into a generic fibre-wall theorem; WR-III owns the geometry. A restriction is an inequality in parameter space before it can be interpreted as any spatial wall.

3 A finite conservation and reversibility interface

Take $X = S = \{0, \dots, n-1\}$ and a single immediate outcome of likelihood one. This is an explicit specialization of the WR-V instrument: $I(r, j \mid i) = P_{ij}$. Thus $P_{ij} \geq 0$ and $\sum_j P_{ij} = 1$. Different

prepared input labels are different rows of one transition model. Assume a known distribution $\pi_i > 0$, $\sum_i \pi_i = 1$, and separately declare detailed balance

$$\pi_i P_{ij} = \pi_j P_{ji}. \quad (2)$$

It implies stationarity by summing over i . This is the standard reversible Markov-chain condition [6, Section 1.6, Proposition 1.20]. Stationarity alone is weaker: a directed three-cycle preserves a uniform distribution but violates (2). We require neither irreducibility nor a physical derivation of an equilibrium bath. Calling a particular physical process reversible would require that extra bridge.

For supplied label energies E_i , exact pathwise conservation is the support condition $P_{ij} = 0$ whenever $E_i \neq E_j$. This condition is physically appropriate only if the labels specify the relevant total energy and the step includes no unmodelled work or exchange. It is not derived from a vanishing mean change.

Proposition 3.1 (Mean balance is not pathwise conservation). *For row-stochastic P and full-support π , pathwise conservation of E is equivalent to $\sum_{i,j} \pi_i P_{ij} (E_j - E_i)^2 = 0$. By contrast, $\pi P = \pi$ always gives $\sum_{i,j} \pi_i P_{ij} (E_j - E_i) = 0$, even when energy-changing transitions occur.*

Proof. Every summand of the squared expression is nonnegative. It vanishes exactly on the required support because $\pi_i > 0$. The mean expression is $(\pi P)E - \pi E$. For a counterexample take $E = (0, 1, 2)$, uniform π and $P_{ij} = 1/3$: the mean change is zero but the squared change is $4/3$. $\square$

Define equilibrium flows $w_{ij} = \pi_i P_{ij}$. For a declared undirected allowed-edge graph G , detailed balance and conservation support give

$$w_{ij} = w_{ji} \geq 0, \quad w_{ij} = 0 \text{ } ((i, j) \notin G, i \neq j), \quad w_{ii} = \pi_i - \sum_{j \neq i} w_{ij} \geq 0. \quad (3)$$

Only edges between equal energies are allowed if exact conservation is imposed. Self-loops are allowed. Independent edge variables determine all rows simultaneously; they are not independent successor simplices. Each inequality and the energy assignment remains part of the model record.

A calibrated binary successor probe has click probabilities $f_j \in [0, 1]$. For repeated preparations of input i its ideal response is

$$q_i = \sum_j P_{ij} f_j = f_i + \frac{1}{\pi_i} \sum_{j \neq i} w_{ij} (f_j - f_i). \quad (4)$$

These probabilities are not a single-run outcome or exact finite-sample frequencies. Equation (4) is affine in the shared edge variables. The inherited WR-V kernel criterion can be applied after eliminating structural equalities; ordinary unconstrained row rank is no longer the whole identification question.

4 Auditable linear feasibility certificates

For finite linear restrictions, vectorize the unknown instrument or use edge coordinates. Collect normalization, structural equations and exact data as $Az = d$, and nonnegative coordinates and bounds as $z \geq 0$, $H z \leq h$. With response tolerance, replace exact response rows by the two appropriate inequalities. This gives standard linear feasibility, not a new optimization principle [7, Sections 2.2.4, 4.3 and 5.8.3].

Proposition 4.1 (A sufficient infeasibility certificate). *If vectors λ and $\mu \geq 0$ satisfy*

$$A^T \lambda + H^T \mu \geq 0, \quad d^T \lambda + h^T \mu < 0, \quad (5)$$

then $\{z \geq 0 : Az = d, H z \leq h\}$ is empty.

Proof. A feasible z would give $0 \leq z^T (A^T \lambda + H^T \mu) = d^T \lambda + \mu^T H z \leq d^T \lambda + \mu^T h < 0$, a contradiction. $\square$

This is a one-way certificate with a direct proof; classical theorems of alternatives give broader converse results under their stated formulations. A numerical solver declaring infeasibility without a verified residual or exact witness is not such a certificate. For a nonempty compact polytope, coordinate minima and maxima identify a singleton exactly when every coordinate range is zero. This elementary diagnostic applies after the existence gate, without interpreting solver precision as exact equality.

Compactness holds for the full row-stochastic carrier with finitely many closed linear restrictions: entries lie in $[0, 1]$. It need not survive a strict requirement such as irreducibility. This distinction becomes decisive at the benchmark threshold.

5 Three states: a complete activity-threshold calculation

Set $n = 3$, $\pi = (1/3, 1/3, 1/3)$ and use all undirected edges. If energy conservation is included, all three labels have the same supplied energy; the probe f below is not energy. Detailed balance gives the symmetric row-stochastic family

$$P(a, b, c) = \begin{pmatrix} 1-a-b & a & b \\ a & 1-a-c & c \\ b & c & 1-b-c \end{pmatrix}, \quad \begin{aligned} & a, b, c \geq 0, \\ & a+b \leq 1, a+c \leq 1, b+c \leq 1. \end{aligned} \quad (6)$$

The parameters a, b, c are transition probabilities, three times the corresponding flows. Repeated preparations of all three inputs and one binary probe with $f = (0, 1/2, 1)$ give

$$q = (a/2 + b, 1/2 - a/2 + c/2, 1 - b - c/2). \quad (7)$$

Fix the exact data $q^* = (1/4, 1/2, 3/4)$. All three prepared-input responses are specified. Their stationary weighted mean is fixed by $\pi P = \pi$, so they do not supply three independent edge constraints.

Theorem 5.1 (An ambiguous reversible completion interval). *The complete feasible class of (6) with responses q^* is*

$$(a, b, c) = (1/6 + 2t, 1/6 - t, 1/6 + 2t), \quad -1/12 \leq t \leq 1/6. \quad (8)$$

Proof. The middle response gives $c = a$. The first gives $b = 1/4 - a/2$; the last then follows. Put $a = 1/6 + 2t$. Nonnegative edges require $t \geq -1/12$ and $t \leq 1/6$. The diagonal entries are $2/3 - t$, $2/3 - 4t$, $2/3 - t$; their nonnegativity adds no stronger bound. Conversely every point of this interval satisfies all constraints and all three responses. $\square$

Define stationary activity as the probability that a step changes its label when the input law is π . Supply an independently justified cap $\nu \geq 0$:

$$\alpha(P) = \sum_i \pi_i \sum_{j \neq i} P_{ij} = \frac{2}{3}(a + b + c) = \frac{1}{3} + 2t, \quad \alpha(P) \leq \nu. \quad (9)$$

This cap is a hard model restriction, not a consequence of q^* or a fitted regularization preference. Minimizing activity without evidence for a hard cap would select a candidate by a decision rule, not establish identifiability.

Theorem 5.2 (Conditional selection and infeasibility threshold). *With (6), exact q^* and the activity cap, the completion class is empty for $\nu < 1/6$, is a singleton for $\nu = 1/6$, and for $\nu > 1/6$ is the nontrivial interval*

$$-1/12 \leq t \leq \min\{1/6, \nu/2 - 1/6\}. \quad (10)$$

The singleton matrix is

$$P_* = P(0, 1/4, 0) = \begin{pmatrix} 3/4 & 0 & 1/4 \\ 0 & 1 & 0 \\ 1/4 & 0 & 3/4 \end{pmatrix}. \quad (11)$$

Proof. Equation (9) converts the cap into $t \leq \nu/2 - 1/6$. Intersect with (8). Its left endpoint is admissible precisely when $\nu \geq 1/6$. Equality leaves that endpoint alone. For a direct infeasibility certificate, $c = a$ and $a/2 + b = 1/4$ imply $\alpha = 1/6 + a \geq 1/6$. Combining this with $\alpha \leq \nu < 1/6$ gives a strict contradiction. This certificate uses declared detailed balance and the exact responses; dropping either changes the problem. $\square$

Corollary 5.3 (An additional strict assumption changes attainment). *If irreducibility is also required, the threshold class is empty at $\nu = 1/6$. For every $\nu > 1/6$ it is nonempty and nonunique. The infimum feasible activity is still $1/6$, but it is not attained.*

Proof. At $t = -1/12$, state 1 is isolated. For every $-1/12 < t \leq 1/6$, the edges $a = c > 0$ connect all states, including the right endpoint where $b = 0$. Excluding the left endpoint turns (10) into a half-open interval whenever $\nu > 1/6$. Its activity values approach $1/6$ from above. $\square$

In the Euclidean topology inherited by (8), the irreducible interval is nonclosed: a minimizing sequence converges to the excluded P_* . This is analogous to WR-II v0.5, Example 10.19 [2], where finite-support profiles approach the excluded all-ones target with residual $1/(N+1)$. The common point is an unattained infimum on a nonclosed admissible set; the carriers and objectives differ.

The same data therefore yield different selection verdicts under different explicit assumptions. This is an audit of the admissibility record, not a paradox or a thermodynamic phase transition. A separately calibrated probe $g = (0, 0, 1)$ at input 0 would measure b and fix $t = 1/6 - b$ on the original interval. Such an observation is different evidence from imposing an activity cap.

6 Response tolerance and uncertain caps

Proposition 6.1 (Positive response tolerance defeats the threshold singleton). *Retain detailed balance and $\nu = 1/6$, but replace exact q^* by $\|Pf - q^*\|_\infty \leq \varepsilon$ for any $\varepsilon > 0$. The compatible class is nonunique.*

Proof. Every $P(0, b, 0)$ with $\max\{0, 1/4 - \varepsilon\} \leq b \leq 1/4$ is row-stochastic and symmetric. Its responses are $(b, 1/2, 1 - b)$ and its activity is $2b/3 \leq 1/6$. The interval has positive length for every $\varepsilon > 0$. It is a contained witness family, not a claim to parameterize the entire tolerance class. $\square$

Next keep exact responses but let the independently supplied cap belong to a closed interval $N = [\nu_-, \nu_+]$. With $\mathcal{C}_\nu = \mathcal{C}_{\nu,0}(q^*)$, define two different classes:

$$\mathcal{C}_\forall(N) = \bigcap_{\nu \in N} \mathcal{C}_\nu, \quad \mathcal{C}_\exists(N) = \bigcup_{\nu \in N} \mathcal{C}_\nu. \quad (12)$$

The first imposes admissibility for every possible cap, a robust design requirement. The second retains a candidate if at least one cap permits it. For inference when the actual cap is one unknown member of N , its joint feasible set is $\{(P, \nu) : \nu \in N, P \in \mathcal{C}_\nu\}$, whose projection onto P is the union, not the intersection. These are hard-set statements, not Bayesian averaging.

Proposition 6.2 (A unique robust candidate need not be uniquely inferred). *For nested scalar upper caps with attained endpoints, $\mathcal{C}_\forall(N) = \mathcal{C}_{\nu_-}$ and $\mathcal{C}_\exists(N) = \mathcal{C}_{\nu_+}$. In the benchmark, take $\nu_- = 1/6$ and $\nu_+ = 1/6 + \delta$ for any $\delta > 0$. The first class is $\{P_*\}$; the second is nonunique.*

Proof. Increasing the cap enlarges the class and both endpoints belong to N . The intersection is its smallest member and the union its largest. Apply Theorem 5.2. $\square$

For a concrete comparison, set $\delta = 1/12$, so $N = [1/6, 1/4]$, and impose no irreducibility requirement. Then $\mathcal{C}_\forall(N) = \{P_*\}$, whereas $\mathcal{C}_\exists(N)$ has $t \in [-1/12, -1/24]$, an interval of length $1/24$. Its right endpoint has $(a, b, c) = (1/12, 5/24, 1/12)$ and $\alpha = 1/4$; its left endpoint is P_* . Both have exactly the same q^* .

For $\nu_- < 1/6 < \nu_+$, robust feasibility fails but possible feasibility remains. That failure does not prove that the unknown actual cap is inconsistent. More generally, simultaneous uncertainty in energies, support graphs, π or calibration matrices need not form a nested family; the endpoint reduction above must not be carried over without proof. Deterministic ε and N have no statistical coverage guarantee until a sampling and calibration model is supplied.

7 From a transition boundary to a physical boundary

The benchmark establishes a boundary of one declared feasibility problem. At $\nu < 1/6$ its exact data cannot be generated by an admissible symmetric matrix with that cap. It does not establish absence of a physical system, because a wrong calibration, a driven nonreversible process, a hidden environment or an inaccurate bound may invalidate the adopted assumptions. Relaxations must be named; they cannot silently count as solutions of the original problem.

The programme's physical categories remain distinct obligations. The following table classifies interfaces to be built, not physical walls already certified here.

Category	Required bridge	Status in this draft
Conservation	Complete conserved quantity and closed-system step.	Conditional finite support rule; not derived from mean balance.
Equilibrium	Known π and independently justified detailed balance.	Classical reversible model; no entropy or dissipation theorem.
Causal	Spacetime/events, durations and operational influence relation.	Open; no speed bound proved.
Quantum	Hilbert spaces, admissible instruments and measured responses.	Open; classical positivity is not complete positivity.
Gravitational	Declared field equations, global conditions and data map.	Open; no cosmological boundary.
Model closure	Audit hidden carriers and allowed relaxations.	Explicit failure diagnostics; no claim of an exhaustive world class.

Even P_* is not a unique complete world. If a declared full carrier is $\mathcal{W} = \{(P, h) : h \in \{0, 1\}\}$ and all present responses and restrictions ignore h , the surviving transition singleton lifts to two full candidates. This is the inherited WR-I/IV projection distinction, not a new hidden-world theorem. A physical extension must define the full carrier, projection, existence conditions and observational meaning of h .

8 Provenance, failure criteria and programme handoff

The proofs use finite positivity, affine elimination, interval intersections and monotonicity of a scalar cap. These established tools have no priority claim. Reversible chains are standard [6], and linear certificate methods are standard [7]. The contribution is an explicit constraint record and a worked transition-completion benchmark that separates observational identification, externally conditioned selection and uncertain-constraint inference. Exhaustive originality of this framework is not certified.

A scientific application must fail its claim gate if an asserted conserved label omits exchanged energy, detailed balance is assumed from stationarity alone, an activity preference is called a measured hard bound, strict constraints are treated as closed, or a robust intersection is confused with projection over an unknown cap. The benchmark can be falsified as a model of an apparatus by independently measured transition probabilities violating its structural assumptions. Such a failure rejects that model, not physical existence.

The next work packages are: (i) independently calibrate a real transition system and justify its constraint record; (ii) build deterministic or statistical uncertainty sets with stated coverage; (iii) construct

one additional causal, quantum or gravitational interface without importing local labels as full worlds. They depend on the declared carrier and evidence, not on numbering alone.

WR-VII may inherit explicitly justified constrained classes to study energy–time frontiers, after assigning physical units and protocols. The dimensionless ν here is not such a frontier. WR-VIII requires real cosmological response maps; WR-IX requires accessible protocol and resource classes. Neither branch is opened or solved by the present finite calculation. WREH remains its own programme and community, distinct from Boundary Compensation.

The supplied exact-arithmetic script verifies matrices, means, activity, threshold cases, irreducibility and tolerance witnesses. The offline bilingual HTML displays the exact-response interval and uncertain-cap quantifiers. Computation illustrates and checks finite witnesses; no theorem uses numerical output as a premise. This v0.2 is REVIEWABLE_DRAFT, with no deposit, DOI, external peer review or journal-readiness claim.

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